SULIT

UTM UNIVERSITI TEKNOLOGI MALAYSIA

**SCHOOL OF COMPUTING**
Faculty of Engineering

# UNIVERSITI TEKNOLOGI MALAYSIA
## FINAL EXAMINATION SEMESTER I, 2018 / 2019

| | | |
|---|---|---|
| **SUBJECT CODE** | : | **SCSI 2143** |
| **SUBJECT NAME** | : | **PROBABILITY & STATISTICAL DATA ANALYSIS** |
| **SECTION** | : | **ALL** |
| **TIME** | : | **9.00 AM – 12.00 NOON** |
| **DATE/DAY** | : | **03 / 01 / 2019 ( THURSDAY )** |
| **VENUES** | : | **BK 2 ( N 28 )** |

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**INSTRUCTIONS :**

Answer **ALL** question in the answer booklet provided.

Provide your final answer accurate to 3 decimal places.

**(Please Write Your Lecturer Name And Section In Your Answer Booklet)**

| | |
|---|---|
| **Name** | |
| **I/C No.** | |
| **Year / Course** | |
| **Section** | |
| **Lecturer Name** | |

This question paper consists of **ELEVEN ( 11 )** printed pages including this page.

**Question 1** **[25 Marks]**

a) Seven housewives were randomly sampled, and it was determined that they walked an average of 39.2 km per week in their housework, with a sample standard deviation of 3.23 km per week. Construct a 95 percent confident interval for the population mean of walked per week of housewives in their housework. (7 marks)

b) Hinton Press hypothesizes that the life of its largest web press is 14,500 hours, with a known standard deviation of 2,800 hours. From a sample of 25 presses, the company finds a sample mean of 13,000 hours. At a 0.01 significance level, should the company conclude that the average life of the presses is less than the hypothesized 14,500 hours? (10 marks)

c) A standard deviation of a distribution is hypothesized to be 50. If an observed sample of 30 yield a sample standard deviation of 57, should we reject the null hypothesis that the true standard deviation is 50? Use the 0.05 level of significance. (8 marks)

**Question 2** **[20 Marks]**

a) A ketchup manufacturer is in the process of deciding whether to produce a new extra spicy brand. The company's marketing-research department used a national telephone survey of 6,000 households and found that the extra-spicy ketchup would be purchased by 335 of them. A much extensive study made 2 years ago showed that 300 of the 6,000 households would purchase the brand then. At a 0.02 significance level, should the company conclude that there is an increased interest in the extra-spicy flavour? (10 marks)

b) A private university has just introduced a new e-learning programme. The university randomly selected 10 students as its research subjects to study whether the programme is effective in improving their performance. The performance score before and after the e-learning program are shown in Table 1:

**Table 1** : Data of student performance

| Student | Performance score before the e-learning programme | Performance score after the e-learning programme |
|---|---|---|
| 1 | 66 | 68 |
| 2 | 34 | 40 |
| 3 | 67 | 66 |
| 4 | 88 | 89 |
| 5 | 56 | 60 |
| 6 | 75 | 76 |
| 7 | 88 | 90 |
| 8 | 66 | 66 |
| 9 | 55 | 58 |
| 10 | 45 | 34 |

i. Determine the effectiveness of the e-learning programme by using a suitable t-test at a 0.05 significance level. (8 marks)

ii. Report the results of the analysis. (2 marks)

**Question 3** **[20 Marks]**

a) A new donut shop plans to sell plain, strawberry, blueberry and cinnamon donuts. They wonder if there is a preference for one of these types of donuts or if each type is preferred by the same proportion of customers. The data of the customer preference is shown in Table 2 :

**Table 2** : Data of customer preference

| | Plain | Strawberry | Blueberry | cinnamon |
|---|---|---|---|---|
| Observed count | 13 | 12 | 16 | 19 |

i. State the null and alternative hypothesis (2 marks)

ii. Find the critical value (3 marks)

iii. Calculate the test statistics (3 marks)

iv. What does the result suggest about the claim that customer preference on each type of donut is similar? (2 marks)

b) A movie theatre recorded the type of snack item purchased and the type of movie the customer was attending for each person in a random sample of theatre customer who made a snack bar purchase. The resulting data as shown in the Table 3. At the 0.05 significant level, test the claim that, type of movies independent with type of snack sold

(10 marks)

**Table 3** : Type of movie with type of snack sold

| | Soda | Popcorn | candy |
|---|---|---|---|
| Children's movie | 70 | 83 | 47 |
| Action movie | 61 | 49 | 20 |

**Question 4** **[20 Marks]**

a) Determine the correlation coefficient for the data in Table 4. Then explain the relationship between thinking skills and IQ based on the correlation coefficient.

(10 marks)

**Table 4** : Data on relationship between thinking skills and IQ

| Subject | Thinking skills score | IQ score |
|---|---|---|
| 1 | 92 | 98 |
| 2 | 60 | 65 |
| 3 | 72 | 78 |
| 4 | 54 | 53 |
| 5 | 88 | 90 |
| 6 | 95 | 100 |
| 7 | 79 | 82 |
| 8 | 72 | 76 |
| 9 | 58 | 58 |
| 10 | 89 | 95 |

b) Zippy Cola is studying the effect of its latest advertising campaign. People chosen at random, were called and asked how many cans of Zippy Cola they had bought in the past week and how many Zippy Cola advertisements they had either read or seen in the past week. The data from their study are presented in Table 5:

**Table 5** : Data on number of advertisements and number of Zipply Cola purchased

| X (number of ads) | 3 | 7 | 4 | 2 | 0 | 4 | 1 | 2 |
|---|---|---|---|---|---|---|---|---|
| Y(can purchased) | 11 | 18 | 9 | 4 | 7 | 6 | 3 | 8 |

Develop the regression equation that best fits the data. (10 marks)

**Question 5** **[15 Marks]**

Students were given different drug treatment before revising for their exam. Some were given a memory drug, some a placebo drug and some no treatment. The exam score (%) are shown below for the three different group. Test the hypotheses that the treatments will have different effect on exam score by using $\alpha = 0.05$. (15 marks)

**Table 6 :** Memory drug for different group

| *n* | Memory drug | placebo | No treatment |
|---|---|---|---|
| 1 | 70 | 37 | 3 |
| 2 | 77 | 43 | 10 |
| 3 | 83 | 50 | 17 |
| 4 | 90 | 57 | 23 |
| 5 | 97 | 63 | 30 |
| mean | 83.4 | 50 | 16.6 |
| variance | 112.3 | 109.0 | 112.3 |

**List of Formula:**

(a) Interval Estimation

$$\bar{x} \pm (Z_{\frac{\alpha}{2}})(\frac{\alpha}{\sqrt{n}})$$

$$\bar{x} \pm (t_{\alpha,dof})(\frac{s}{\sqrt{n}})$$

(b) Hypothesis Testing for one sample

$$Test\ Statistic, z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$Test\ Statistic, t_0 = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$Test\ Statistic, \chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

(c) Hypothesis Testing for two samples

$$Test\ Statistic, z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\bar{p}\,\bar{q}}{n_1} + \frac{\bar{p}\,\bar{q}}{n_2}}}$$

With $\hat{p}_1 = \frac{x_1}{n_1}$, $\hat{p}_2 = \frac{x_2}{n_2}$, $\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$, $\bar{q} = 1 - \bar{p}$

$$Test\ Statistic, z_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$Test\ Statistic, t_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}; s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$Test\ Statistic, T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}; v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

$$Test\ Statistic, T_0 = \frac{\bar{D} - u_D}{\frac{S_D}{\sqrt{n}}}$$

$$Test\ Statistic, F_0 = \frac{s_1^2}{s_2^2}$$

$$S_D = \sqrt{\frac{\sum D^2 - \frac{\sum D^2}{n}}{n-1}}$$

(d) Chi-Square Test

$$Test\ Statistic, \chi^2 = \sum_{i}^{n} \frac{(o_i - e_i)^2}{e_i}$$

$$Degree\ of\ Freedom\ for\ two-way\ contigency\ table = (r-1)(c-1)$$

(e) Correlation

$$Pearson\ Correlation\ Coefficient, r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)\left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$Spearman\ Correlation\ Coefficient, r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}\ where\ d_i = y_i - x_i$$

(f) Regression

$$b_1 = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

(g) Analysis of variance (ANOVA)

$$Test\ Statistic, F = \frac{ns_{\bar{x}}^2}{s_p^2}$$

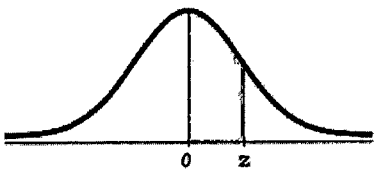



**TABLE A-2** Standard Normal (z) Distribution

| z | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | .0000 | .0040 | .0080 | .0120 | .0160 | .0199 | .0239 | .0279 | .0319 | .0359 |
| 0.1 | .0398 | .0438 | .0478 | .0517 | .0557 | .0596 | .0636 | .0675 | .0714 | .0753 |
| 0.2 | .0793 | .0832 | .0871 | .0910 | .0948 | .0987 | .1026 | .1064 | .1103 | .1141 |
| 0.3 | .1179 | .1217 | .1255 | .1293 | .1331 | .1368 | .1406 | .1443 | .1480 | .1517 |
| 0.4 | .1554 | .1591 | .1628 | .1664 | .1700 | .1736 | .1772 | .1808 | .1844 | .1879 |
| 0.5 | .1915 | .1950 | .1985 | .2019 | .2054 | .2088 | .2123 | .2157 | .2190 | .2224 |
| 0.6 | .2257 | .2291 | .2324 | .2357 | .2389 | .2422 | .2454 | .2486 | .2517 | .2549 |
| 0.7 | .2580 | .2611 | .2642 | .2673 | .2704 | .2734 | .2764 | .2794 | .2823 | .2852 |
| 0.8 | .2881 | .2910 | .2939 | .2967 | .2995 | .3023 | .3051 | .3078 | .3106 | .3133 |
| 0.9 | .3159 | .3186 | .3212 | .3238 | .3264 | .3289 | .3315 | .3340 | .3365 | .3389 |
| 1.0 | .3413 | .3438 | .3461 | .3485 | .3508 | .3531 | .3554 | .3577 | .3599 | .3621 |
| 1.1 | .3643 | .3665 | .3686 | .3708 | .3729 | .3749 | .3770 | .3790 | .3810 | .3830 |
| 1.2 | .3849 | .3869 | .3888 | .3907 | .3925 | .3944 | .3962 | .3980 | .3997 | .4015 |
| 1.3 | .4032 | .4049 | .4066 | .4082 | .4099 | .4115 | .4131 | .4147 | .4162 | .4177 |
| 1.4 | .4192 | .4207 | .4222 | .4236 | .4251 | .4265 | .4279 | .4292 | .4306 | .4319 |
| 1.5 | .4332 | .4345 | .4357 | .4370 | .4382 | .4394 | .4406 | .4418 | .4429 | .4441 |
| 1.6 | .4452 | .4463 | .4474 | .4484 | .4495 * | .4505 | .4515 | .4525 | .4535 | .4545 |
| 1.7 | .4554 | .4564 | .4573 | .4582 | .4591 | .4599 | .4608 | .4616 | .4625 | .4633 |
| 1.8 | .4641 | .4649 | .4656 | .4664 | .4671 | .4678 | .4686 | .4693 | .4699 | .4706 |
| 1.9 | .4713 | .4719 | .4726 | .4732 | .4738 | .4744 | .4750 | .4756 | .4761 | .4767 |
| 2.0 | .4772 | .4778 | .4783 | .4788 | .4793 | .4798 | .4803 | .4808 | .4812 | .4817 |
| 2.1 | .4821 | .4826 | .4830 | .4834 | .4838 | .4842 | .4846 | .4850 | .4854 | .4857 |
| 2.2 | .4861 | .4864 | .4868 | .4871 | .4875 | .4878 | .4881 | .4884 | .4887 | .4890 |
| 2.3 | .4893 | .4896 | .4898 | .4901 | .4904 | .4906 | .4909 | .4911 | .4913 | .4916 |
| 2.4 | .4918 | .4920 | .4922 | .4925 | .4927 | .4929 | .4931 | .4932 | .4934 | .4936 |
| 2.5 | .4938 | .4940 | .4941 | .4943 | .4945 | .4946 | .4948 | .4949 * | .4951 | .4952 |
| 2.6 | .4953 | .4955 | .4956 | .4957 | .4959 | .4960 | .4961 | .4962 | .4963 | .4964 |
| 2.7 | .4965 | .4966 | .4967 | .4968 | .4969 | .4970 | .4971 | .4972 | .4973 | .4974 |
| 2.8 | .4974 | .4975 | .4976 | .4977 | .4977 | .4978 | .4979 | .4979 | .4980 | .4981 |
| 2.9 | .4981 | .4982 | .4982 | .4983 | .4984 | .4984 | .4985 | .4985 | .4986 | .4986 |
| 3.0 | .4987 | .4987 | .4987 | .4988 | .4988 | .4989 | .4989 | .4989 | .4990 | .4990 |
| 3.10 and higher | .4999 | | | | | | | | | |

*NOTE:* For values of z above 3.09, use 0.4999 for the area.

*Use these common values that result from interpolation:

| z score | Area |
|---|---|
| 1.645 | 0.4500 |
| 2.575 | 0.4950 |

From Frederick C. Mosteller and Robert E. K. Rourke, *Sturdy Statistics*, 1973, Addison-Wesley Publishing Co., Reading, MA. Reprinted with permission of Frederick Mosteller.

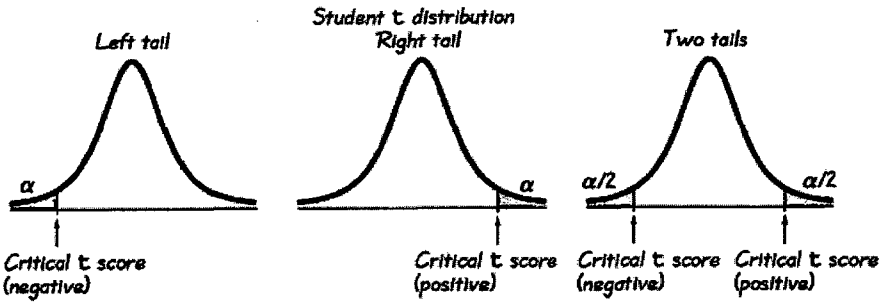



**TABLE A-3** t Distribution

| | α | | | | | |
|---|---|---|---|---|---|---|
| Degrees of Freedom | .005 (one tail) .01 (two tails) | .01 (one tail) .02 (two tails) | .025 (one tail) .05 (two tails) | .05 (one tail) .10 (two tails) | .10 (one tail) .20 (two tails) | .25 (one tail) .50 (two tails) |
| 1 | 63.657 | 31.821 | 12.706 | 6.314 | 3.078 | 1.000 |
| 2 | 9.925 | 6.965 | 4.303 | 2.920 | 1.886 | .816 |
| 3 | 5.841 | 4.541 | 3.182 | 2.353 | 1.638 | .765 |
| 4 | 4.604 | 3.747 | 2.776 | 2.132 | 1.533 | .741 |
| 5 | 4.032 | 3.365 | 2.571 | 2.015 | 1.476 | .727 |
| 6 | 3.707 | 3.143 | 2.447 | 1.943 | 1.440 | .718 |
| 7 | 3.500 | 2.998 | 2.365 | 1.895 | 1.415 | .711 |
| 8 | 3.355 | 2.896 | 2.306 | 1.860 | 1.397 | .706 |
| 9 | 3.250 | 2.821 | 2.262 | 1.833 | 1.383 | .703 |
| 10 | 3.169 | 2.764 | 2.228 | 1.812 | 1.372 | .700 |
| 11 | 3.106 | 2.718 | 2.201 | 1.796 | 1.363 | .697 |
| 12 | 3.054 | 2.681 | 2.179 | 1.782 | 1.356 | .696 |
| 13 | 3.012 | 2.650 | 2.160 | 1.771 | 1.350 | .694 |
| 14 | 2.977 | 2.625 | 2.145 | 1.761 | 1.345 | .692 |
| 15 | 2.947 | 2.602 | 2.132 | 1.753 | 1.341 | .691 |
| 16 | 2.921 | 2.584 | 2.120 | 1.746 | 1.337 | .690 |
| 17 | 2.898 | 2.567 | 2.110 | 1.740 | 1.333 | .689 |
| 18 | 2.878 | 2.552 | 2.101 | 1.734 | 1.330 | .688 |
| 19 | 2.861 | 2.540 | 2.093 | 1.729 | 1.328 | .688 |
| 20 | 2.845 | 2.528 | 2.086 | 1.725 | 1.325 | .687 |
| 21 | 2.831 | 2.518 | 2.080 | 1.721 | 1.323 | .686 |
| 22 | 2.819 | 2.508 | 2.074 | 1.717 | 1.321 | .686 |
| 23 | 2.807 | 2.500 | 2.069 | 1.714 | 1.320 | .685 |
| 24 | 2.797 | 2.492 | 2.064 | 1.711 | 1.318 | .685 |
| 25 | 2.787 | 2.485 | 2.060 | 1.708 | 1.316 | .684 |
| 26 | 2.779 | 2.479 | 2.056 | 1.706 | 1.315 | .684 |
| 27 | 2.771 | 2.473 | 2.052 | 1.703 | 1.314 | .684 |
| 28 | 2.763 | 2.467 | 2.048 | 1.701 | 1.313 | .683 |
| 29 | 2.756 | 2.462 | 2.045 | 1.699 | 1.311 | .683 |
| Large (z) | 2.575 | 2.326 | 1.960 | 1.645 | 1.282 | .675 |

**TABLE A-4** Chi-Square ($\chi^2$) Distribution

| Degrees of Freedom | Area to the Right of the Critical Value | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 0.995 | 0.99 | 0.975 | 0.95 | 0.90 | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
| 1 | — | — | 0.001 | 0.004 | 0.016 | 2.706 | 3.841 | 5.024 | 6.635 | 7.879 |
| 2 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 3 | 0.072 | 0.115 | 0.216 | 0.352 | 0.584 | 6.251 | 7.815 | 9.348 | 11.345 | 12.838 |
| 4 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 5 | 0.412 | 0.554 | 0.831 | 1.145 | 1.610 | 9.236 | 11.071 | 12.833 | 15.086 | 16.750 |
| 6 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 7 | 0.989 | 1.239 | 1.690 | 2.167 | 2.833 | 12.017 | 14.067 | 16.013 | 18.475 | 20.278 |
| 8 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 9 | 1.735 | 2.088 | 2.700 | 3.325 | 4.168 | 14.684 | 16.919 | 19.023 | 21.666 | 23.589 |
| 10 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 11 | 2.603 | 3.053 | 3.816 | 4.575 | 5.578 | 17.275 | 19.675 | 21.920 | 24.725 | 26.757 |
| 12 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 13 | 3.565 | 4.107 | 5.009 | 5.892 | 7.042 | 19.812 | 22.362 | 24.736 | 27.688 | 29.819 |
| 14 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 15 | 4.601 | 5.229 | 6.262 | 7.261 | 8.547 | 22.307 | 24.996 | 27.488 | 30.578 | 32.801 |
| 16 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 17 | 5.697 | 6.408 | 7.564 | 8.672 | 10.085 | 24.769 | 27.587 | 30.191 | 33.409 | 35.718 |
| 18 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 19 | 6.844 | 7.633 | 8.907 | 10.117 | 11.651 | 27.204 | 30.144 | 32.852 | 36.191 | 38.582 |
| 20 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 21 | 8.034 | 8.897 | 10.283 | 11.591 | 13.240 | 29.615 | 32.671 | 35.479 | 38.932 | 41.401 |
| 22 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 23 | 9.260 | 10.196 | 11.689 | 13.091 | 14.848 | 32.007 | 35.172 | 38.076 | 41.638 | 44.181 |
| 24 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 25 | 10.520 | 11.524 | 13.120 | 14.611 | 16.473 | 34.382 | 37.652 | 40.646 | 44.314 | 46.928 |
| 26 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 27 | 11.808 | 12.879 | 14.573 | 16.151 | 18.114 | 36.741 | 40.113 | 43.194 | 46.963 | 49.645 |
| 28 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 29 | 13.121 | 14.257 | 16.047 | 17.708 | 19.768 | 39.087 | 42.557 | 45.722 | 49.588 | 52.336 |
| 30 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 40 | 20.707 | 22.164 | 24.433 | 26.509 | 29.051 | 51.805 | 55.758 | 59.342 | 63.691 | 66.766 |
| 50 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 60 | 35.534 | 37.485 | 40.482 | 43.188 | 46.459 | 74.397 | 79.082 | 83.298 | 88.379 | 91.952 |
| 70 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 80 | 51.172 | 53.540 | 57.153 | 60.391 | 64.278 | 96.578 | 101.879 | 106.629 | 112.329 | 116.321 |
| 90 | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 100 | 67.328 | 70.065 | 74.222 | 77.929 | 82.358 | 118.498 | 124.342 | 129.561 | 135.807 | 140.169 |

F Values for $\alpha = 0.05$

| $d_2$ | $d_1$ 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 161.4 | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 240.5 |
| 2 | 18.51 | 19.00 | 19.16 | 19.25 | 19.3 | 19.33 | 19.35 | 19.37 | 19.38 |
| 3 | 10.13 | 9.55 | 9.28 | 9.12 | 9.01 | 8.94 | 8.89 | 8.85 | 8.81 |
| 4 | 7.71 | 6.94 | 6.59 | 6.39 | 6.26 | 6.16 | 6.09 | 6.04 | 6.00 |
| 5 | 6.61 | 5.79 | 5.41 | 5.19 | 5.05 | 4.95 | 4.88 | 4.82 | 4.77 |
| 6 | 5.99 | 5.14 | 4.76 | 4.53 | 4.39 | 4.28 | 4.21 | 4.15 | 4.10 |
| 7 | 5.59 | 4.74 | 4.35 | 4.12 | 3.97 | 3.87 | 3.79 | 3.73 | 3.68 |
| 8 | 5.32 | 4.46 | 4.07 | 3.84 | 3.69 | 3.58 | 3.50 | 3.44 | 3.39 |
| 9 | 5.12 | 4.26 | 3.86 | 3.63 | 3.48 | 3.37 | 3.29 | 3.23 | 3.18 |
| 10 | 4.96 | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 |
| 11 | 4.84 | 3.98 | 3.59 | 3.36 | 3.20 | 3.09 | 3.01 | 2.95 | 2.90 |
| 12 | 4.75 | 3.89 | 3.49 | 3.26 | 3.11 | 3.00 | 2.91 | 2.85 | 2.80 |
| 13 | 4.67 | 3.81 | 3.41 | 3.18 | 3.03 | 2.92 | 2.83 | 2.77 | 2.71 |
| 14 | 4.60 | 3.74 | 3.34 | 3.11 | 2.96 | 2.85 | 2.76 | 2.70 | 2.65 |
| 15 | 4.54 | 3.68 | 3.29 | 3.06 | 2.90 | 2.79 | 2.71 | 2.64 | 2.59 |
| 16 | 4.49 | 3.63 | 3.24 | 3.01 | 2.85 | 2.74 | 2.66 | 2.59 | 2.54 |
| 17 | 4.45 | 3.59 | 3.20 | 2.96 | 2.81 | 2.70 | 2.61 | 2.55 | 2.49 |
| 18 | 4.41 | 3.55 | 3.16 | 2.93 | 2.77 | 2.66 | 2.58 | 2.51 | 2.46 |
| 19 | 4.38 | 3.52 | 3.13 | 2.90 | 2.74 | 2.63 | 2.54 | 2.48 | 2.42 |
| 20 | 4.35 | 3.49 | 3.10 | 2.87 | 2.71 | 2.60 | 2.51 | 2.45 | 2.39 |
| 21 | 4.32 | 3.47 | 3.07 | 2.84 | 2.68 | 2.57 | 2.49 | 2.42 | 2.37 |
| 22 | 4.30 | 3.44 | 3.05 | 2.82 | 2.66 | 2.55 | 2.46 | 2.40 | 2.34 |
| 23 | 4.28 | 3.42 | 3.03 | 2.80 | 2.64 | 2.53 | 2.44 | 2.37 | 2.32 |
| 24 | 4.26 | 3.40 | 3.01 | 2.78 | 2.62 | 2.51 | 2.42 | 2.36 | 2.30 |
| 25 | 4.24 | 3.39 | 2.99 | 2.76 | 2.60 | 2.49 | 2.40 | 2.34 | 2.28 |
| 26 | 4.23 | 3.37 | 2.98 | 2.74 | 2.59 | 2.47 | 2.39 | 2.32 | 2.27 |
| 27 | 4.21 | 3.35 | 2.96 | 2.73 | 2.57 | 2.46 | 2.37 | 2.31 | 2.25 |
| 28 | 4.20 | 3.34 | 2.95 | 2.71 | 2.56 | 2.45 | 2.36 | 2.29 | 2.24 |
| 29 | 4.18 | 3.33 | 2.93 | 2.70 | 2.55 | 2.43 | 2.35 | 2.28 | 2.22 |
| 30 | 4.17 | 3.32 | 2.92 | 2.69 | 2.53 | 2.42 | 2.33 | 2.27 | 2.21 |
| 40 | 4.08 | 3.23 | 2.84 | 2.61 | 2.45 | 2.34 | 2.25 | 2.18 | 2.12 |
| 60 | 4.00 | 3.15 | 2.76 | 2.53 | 2.37 | 2.25 | 2.17 | 2.10 | 2.04 |
| 120 | 3.92 | 3.07 | 2.68 | 2.45 | 2.29 | 2.17 | 2.09 | 2.02 | 1.96 |
| inf | 3.84 | 3.00 | 2.60 | 2.37 | 2.21 | 2.10 | 2.01 | 1.94 | 1.88 |